Policy Gradient Method

Deep Reinforcement Learning (DRL)

Policy Gradient Method

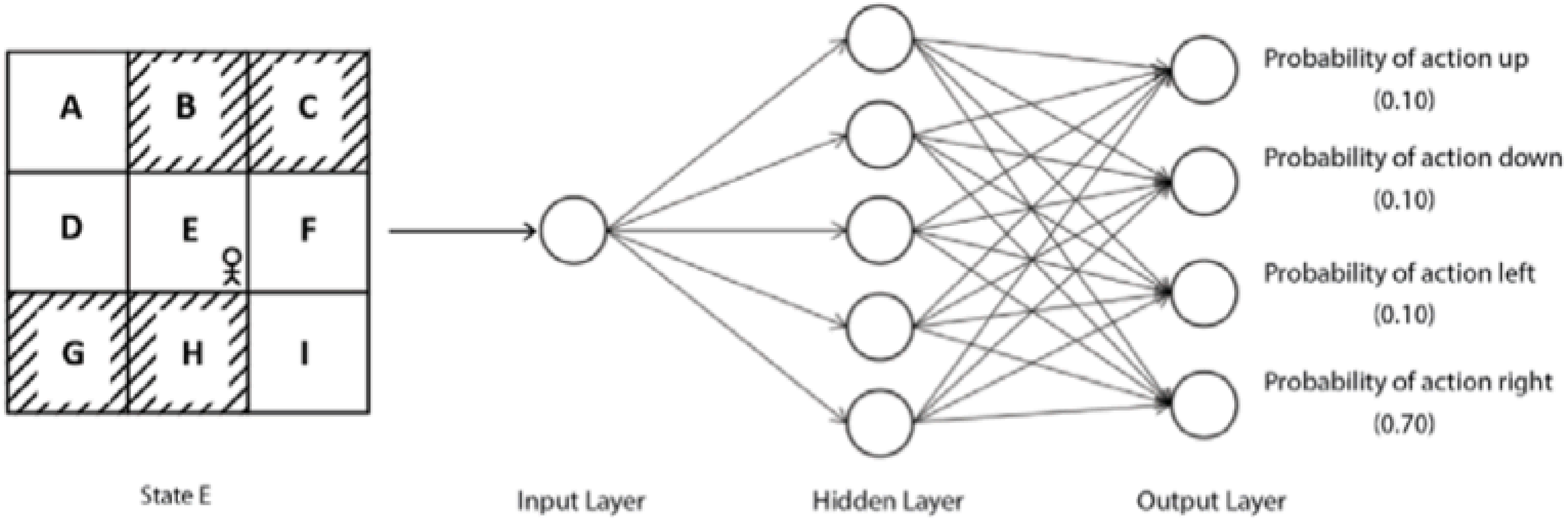
Value-based methods involve iteratively calculating the optimal Q function and deriving the optimal policy from it. However, in this chapter, we explore policy-based methods, which allow us to determine the optimal policy without the need to compute the optimal Q function.

Why policy-based methods?

- Many real-world challenges involve continuous action spaces, such as self-driving cars or robots learning to walk, which are not only continuous but also high-dimensional.
- Methods like DQN and other value-based approaches struggle to handle continuous action spaces effectively.
- Policy-based methods offer a solution by directly computing the optimal policy without the need to calculate Q values.
- This means extracting the policy without relying on the Q function.
- Policy-based methods offer various benefits over value-based methods and are capable of managing both discrete and continuous action spaces.

Policy gradient intuition

- Policy gradient stands out as a highly favored algorithm within deep reinforcement learning. This method, known as a policy-based approach, allows us to discover the optimal policy without the need to compute the Q function. Instead, it achieves this by directly parameterizing the policy using a specific parameter θ.
- In utilizing the policy gradient method, a stochastic policy comes into play. This type of policy involves selecting actions based on the probability distribution across the action space. For instance, if we have a stochastic policy π, it provides the likelihood of taking an action a given a state s, denoted as $\pi(a|s)$. By employing a parameterized policy within the policy gradient method, we can represent our policy as $\pi(a|s;\theta)$, showcasing that our policy is parameterized.



- When we feed the state E as an input to the network, it will return the probability distribution over all actions in our action space.
- In the policy gradient method, the network returns the probability distribution (action probabilities) over the action space

Algorithm – policy gradient

The policy gradient algorithm is used in reinforcement learning to find the optimal policy directly, without explicitly computing the value function.

1. **Policy Representation**: The policy is typically represented as a parameterized function, such as a neural network, that takes the state as input and outputs the probability distribution over actions.
2. **Objective Function**: The objective of the policy gradient algorithm is to maximize the expected cumulative reward, which is often expressed as the expected value of the sum of rewards over time steps.
3. **Gradient Ascent:** The algorithm uses gradient ascent to update the policy parameters, moving them in the direction that increases the expected cumulative reward.
4. **Policy Gradient Theorem**: The policy gradient theorem provides a way to compute the gradient of the expected cumulative reward with respect to the policy parameters.
5. **REINFORCE Algorithm**: The REINFORCE algorithm is a specific instance of the policy gradient algorithm that uses Monte Carlo estimation to compute the gradient.
6. **Advantages and Disadvantages**: Policy gradient methods can handle continuous action spaces and are often more sample-efficient than value-based methods like Q-learning. However, they can be sensitive to the choice of hyperparameters and require careful tuning.
7. **Variants**: There are several variants of the policy gradient algorithm, such as actor-critic methods, which use a value function to reduce the variance of the gradient estimates.
8. Exploration vs. Exploitation: Like other reinforcement learning algorithms, policy gradient methods must balance exploration (trying new actions) and exploitation (using the current best action) to find the optimal policy.

Variance reduction methods

The basic policy gradient technique, known as the REINFORCE method, faces a significant challenge due to the high variance in the gradient ($\nabla J(\theta)$) during updates. This variance primarily stems from variations in episodic returns. Policy gradient is an on-policy method, meaning we enhance the same policy used to generate episodes continuously.

As a result, the returns fluctuate significantly in each episode, leading to high variance in gradient updates. When gradients display high variance, achieving convergence takes a considerable amount of time.

To reduce the variance we could use:

- Policy gradients with reward-to-go (causality)
- Policy gradients with baseline

Cart pole balancing with policy gradient

Lab

Summary

- Value-based methods involve deriving the best policy from the optimal Q function (Q values). However, calculating the Q function becomes challenging when the action space is continuous. While one option is to discretize the action space, this approach may not always be ideal as it can result in the loss of key features and a large number of actions.
- The policy-based method allows us to determine the optimal policy without relying on the Q function. One prominent technique within the policy-based approach is the policy gradient method, where the optimal policy is directly identified by parameterizing the policy with a parameter θ.
- In the policy gradient method, actions are chosen based on the probability distribution of actions provided by the network. If an episode is successful, meaning a high return is achieved, the probabilities of all actions in the episode are increased; conversely, they are reduced if the episode is not successful. We then delved into the step-by-step derivation of the policy gradient and explored the algorithm of the policy gradient method in more depth.
- Variance reduction techniques can be used like reward-to-go and the policy gradient method with a baseline function.
- In the policy gradient method with a baseline function, two networks are employed: the policy network for determining the optimal policy and the value network for adjusting the gradient

Actor-Critic Methods – A2C and A3C

Actor-Critic method

1. The actor-critic method is one of the most popular algorithms in deep reinforcement learning. Several modern deep reinforcement learning algorithms are designed based on actor-critic methods. The actor-critic method lies at the intersection of value- based and policy-based methods. That is, it takes advantage of both value-based and policy-based methods.
2. Actor-critic, as the name suggests, consists of two types of network—the actor network and the critic network. The role of the actor network is to find an optimal policy, while the role of the critic network is to evaluate the policy produced by the actor network.

Actor-critic method

- The actor network is basically the policy network, and it finds the optimal policy using a policy gradient method. The critic network is basically the value network, and it estimates the state value.
- Using its state value, the critic network evaluates the action produced by the actor network and sends its feedback to the actor. Based on the critic's feedback, the actor network then updates its parameter.
- Thus, in the actor-critic method, we use two networks—the actor network (policy network), which computes the policy, and the critic network (value network), which evaluates the policy produced by the actor network by computing the value function (state values).

Actor-critic method

- The REINFORCE with baseline technique reveals the presence of two networks: the policy network, responsible for identifying the optimal policy, and the value network, which acts as a baseline to regulate the variance in gradient updates. Similarly, the actor-critic method consists of an actor network (policy network) and a critic network (value network), mirroring the setup of REINFORCE with baseline.
- A key contrast between the REINFORCE with baseline method and the actor-critic method lies in the timing of network parameter updates. In the REINFORCE with baseline method, updates occur at the end of an episode, whereas in the actor-critic method, updates happen at each step of the episode.

The actor-critic algorithm

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Mountain car climbing using A3C

Thank you